

### Foundation (Jalapeno)

- Q1.** A basketball player scores  $2 \cdot \sqrt{2}$  points per game. Rationalize the denominator of  $\frac{1}{2 \cdot \sqrt{2}}$ .
- (A)  $\frac{\sqrt{2}}{4}$   
(B)  $\frac{1}{\sqrt{2}}$   
(C)  $\frac{\sqrt{2}}{2}$   
(D)  $\frac{1}{2 \cdot \sqrt{2}}$   
(E)  $\frac{2}{\sqrt{2}}$
- Q2.** A cyclist travels  $5 \cdot \sqrt{3}$  miles in 1 hour. Rationalize the denominator of  $\frac{1}{5 \cdot \sqrt{3}}$ .
- (A)  $\frac{\sqrt{3}}{15}$   
(B)  $\frac{1}{\sqrt{3}}$   
(C)  $\frac{\sqrt{3}}{5}$   
(D)  $\frac{1}{5 \cdot \sqrt{3}}$   
(E)  $\frac{5}{\sqrt{3}}$
- Q3.** The distance between two points on a soccer field is  $3 \cdot \sqrt{2}$  meters. Rationalize the denominator of  $\frac{1}{3 \cdot \sqrt{2}}$ .
- (A)  $\frac{\sqrt{2}}{6}$   
(B)  $\frac{1}{\sqrt{2}}$   
(C)  $\frac{\sqrt{2}}{3}$   
(D)  $\frac{1}{3 \cdot \sqrt{2}}$   
(E)  $\frac{3}{\sqrt{2}}$
- Q4.** A gymnast scores  $4 \cdot \sqrt{5}$  points for a routine. Rationalize the denominator of  $\frac{1}{4 \cdot \sqrt{5}}$ .
- (A)  $\frac{\sqrt{5}}{20}$   
(B)  $\frac{1}{\sqrt{5}}$   
(C)  $\frac{\sqrt{5}}{4}$   
(D)  $\frac{1}{4 \cdot \sqrt{5}}$   
(E)  $\frac{4}{\sqrt{5}}$
- Q5.** A tennis player hits  $2 \cdot \sqrt{6}$  aces per set. Rationalize the denominator of  $\frac{1}{2 \cdot \sqrt{6}}$ .
- (A)  $\frac{\sqrt{6}}{12}$   
(B)  $\frac{1}{\sqrt{6}}$   
(C)  $\frac{\sqrt{6}}{2}$   
(D)  $\frac{1}{2 \cdot \sqrt{6}}$   
(E)  $\frac{2}{\sqrt{6}}$
- Q6.** The length of a hockey rink is  $10 \cdot \sqrt{2}$  meters. Rationalize the denominator of  $\frac{1}{10 \cdot \sqrt{2}}$ .
- (A)  $\frac{\sqrt{2}}{20}$   
(B)  $\frac{1}{\sqrt{2}}$   
(C)  $\frac{\sqrt{2}}{10}$   
(D)  $\frac{1}{10 \cdot \sqrt{2}}$   
(E)  $\frac{10}{\sqrt{2}}$
- Q7.** A volleyball team wins  $6 \cdot \sqrt{3}$  games in a season. Rationalize the denominator of  $\frac{1}{6 \cdot \sqrt{3}}$ .
- (A)  $\frac{\sqrt{3}}{18}$   
(B)  $\frac{1}{\sqrt{3}}$   
(C)  $\frac{\sqrt{3}}{6}$   
(D)  $\frac{1}{6 \cdot \sqrt{3}}$   
(E)  $\frac{6}{\sqrt{3}}$
- Q8.** The width of a basketball court is  $5 \cdot \sqrt{5}$  meters. Rationalize the denominator of  $\frac{1}{5 \cdot \sqrt{5}}$ .
- (A)  $\frac{\sqrt{5}}{25}$   
(B)  $\frac{1}{\sqrt{5}}$   
(C)  $\frac{\sqrt{5}}{5}$   
(D)  $\frac{1}{5 \cdot \sqrt{5}}$   
(E)  $\frac{5}{\sqrt{5}}$

#### Open Ended Questions (FRQ)

- Q9.** A runner travels  $8 \cdot \sqrt{2}$  kilometers in a race. Rationalize the denominator of  $\frac{1}{8 \cdot \sqrt{2}}$ . Show all steps and explain your reasoning.
- Q10.** A swimmer completes  $9 \cdot \sqrt{3}$  laps in a competition. Rationalize the denominator of  $\frac{1}{9 \cdot \sqrt{3}}$ . Use mathematical notation to show your work.

#### Chef's Tips

- Chef's Tips: Ensure students show all steps and reasoning in free response questions.
- Chef's Tips: Encourage students to check their answers by plugging them back into the original equation.
- Chef's Tips: Remind students to simplify their answers and rationalize denominators where necessary.